5) اكتب برنامج لطباعة ~~[illegible]~~ المثلث الأعلى وبدلها برقم 1

$$\begin{bmatrix} 2 & 3 & 4 \\ 5 & 0 & 6 \\ 4 & 2 & 3 \end{bmatrix}$$

Q2> Identify and correct the errors of the following set of code then print out the expected output: **(3 marks)**

```
String str = java language ;
System.out.println(str.lngth());
System.out.println(str.charat(4));
System.out.println(str.trim);
System.out.println(str.toUpperCase());
System.out.println(str ());
```

Group A: Write a suitable Java program to merge two arrays into a third.
For example: A= {2, 4}, B= {3, 6, 8} C= {2, 4, 3, 6, 8}

.....

Group C: Write a Java program to count prime numbers in an 1-D array.

B- Fill in the blanks in each of the following statements (**Answer 4 only**):

(4 marks)

1- The action of changing a data type into a different type is called _____.

2- The indexes of the array start with _____ always.

3- Method name _____compares two strings. It returns true if these are identical, otherwise return false.

4- next() is a type of _____ methods.

5- The _____ statement terminates the current iteration of a loop and proceeds directly to the next instruction.

Q2 Fill in the blanks (5 marks)

1. The ---------- usually used with the switch-case statement; however, it can be used with a loop statement where it terminates the loop.
2. the string index begins at ----------.
3. length Returns the length of the string as ----------.
4. int a = (int)18.7; a=----------
5. A ---------- is an indexed list of values that all have the same type.

Q2> Fill in the blanks in each of the following statements: **(5 marks)**

1. The action of changing a data type into a different type is called _____.
2. The indexes of the array start with _____ always.
3. Method name _____ compares two strings. It returns true if these are identical, otherwise return false.
4. Array is a _____.
5. nextInt() is a type of _____ methods.

Q2 Fill in the blanks (5 marks)

1. In Java, the class --------- provides pre-written mathematical functions to perform various math calculations.
2. The method ---------- used to find the natural logarithm of x (base e).
3. System.out.println(Math.ceil(45.87)); output is----------
4. The method used to compare two strings without considering the case of small or capital letters is ----------
5. The ---------- statement skip the current iteration of a loop and proceeds directly to the next iteration.

B- Fill in the blanks in each of the following statements (**Answer 4 only**):

(4 marks)

1- The action of changing a data type into a different type is called _____.

2- The indexes of the array start with _____ always.

3- Method name _____compares two strings. It returns true if these are identical, otherwise return false.

4- next() is a type of _____ methods.

5- The _____ statement terminates the current iteration of a loop and proceeds directly to the next instruction.

Q1> Determine whether each of the following is true or false: **(5 marks)**

1- If you create an array A= new int [3] you should think of it as a "*Array*" with 3 rows and 1 columns.
2- ceil(X) Rounds X to the largest integer not greater than X.
3- To access the elements of an array we use the { } operator.
4- Java is not an OOP (Object Oriented Programming) Language.
5- You can process all the elements of an array via a loop.

(5 marks)

Note: Attempt all questions

Q1> A- *Determine whether each of the following is true or false* **(Answer 4 only):**

(4 marks)

1- *If you create an array A= new int [5][6], you should think of it as a "matrix" with 6 rows and 4 columns.*

2- *exp(x): the natural logarithm of x.*

3- *To access the elements of an array we use the { } operator.*

4- *Java is not an OOP (Object Oriented Programming) Language.*

5- *You can process all the elements of an array via the switch-case statement.*

Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextInt() method, which is used to read string. F
2. floor(x) method Rounds x to the smallest integer not less than x? F
3. To access the elements of an array we use the { } operator. T
4. equals method Removes any spaces at the beginning or end of a string. F
5. The output of charAt(i) method is int. F

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Note: Attempt all questions

Q1> A- Determine whether each of the following is true or false (**Answer 4 only**): **(4 marks)**

1- If you create an array A= new int [5][6], you should think of it as a "*matrix*" with 6 rows and 4 columns.

2- exp(x): the natural logarithm of x.

3- To access the elements of an array we use the { } operator.

4- Java is not an OOP (Object Oriented Programming) Language.

5- You can process all the elements of an array via the switch-case statement.

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Course Stage / Level : 1st
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Time : 1 Hours

Note: Answer all questions.

Q1 TRUE OR FALSE (5 marks)

1. The nextLine() method, which is used to read int.
2. ceil(x) method Rounds x to the largest integer not greater than x
1. To access the elements of an array we use the [] operator.
2. toUpperCase method Returns a copy of the original string, all lower case.
3. The output of Substring(i,j) method is int.

Q3> Identify and correct the errors of the following set of code (**Answer 2 only**):

(6 marks)

1)
```
int a=11.5; b=5.3;
System.out.println(math.Max(a, b));
System.out.println(math.abs(a));
```

2)
```
char ch = 'A';
int num = ch;
System.out.print(nume + cher);
```

3)
```
printSquare("hello");
printSquare(3.5)
public static void printSquare(int x) {
System.out.println(x*x);
}
```

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Course Name: Advanced Programming

Course Stage/Level: 1st
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Time: Two hours

Q2> Find the output of the following code (**Answer 2 only**): **(6 marks)**

1)
```
for (i = 1; i <= 9; i++ )
{
    for (j = 1; j <= i; j++ )
    {
        System.out.print((i+1)+" ");
    }
System.out.println();
}
```

2)
```
start = "Talk is cheap. ";
middle = "Show me the code. ";
end = "- Linus Torvalds";
result = start + middle + end;
System.out.println(result);
System.out.println(result.charAt(7));
```

3)
```
int[] Array = {2, 5, 1, 9, -11};
System.out.println(Array.length);
System.out.println(Array[-2]);
```

Q4> Identify and correct the errors of the following set of code: (6 marks)

1)
```
Scanner A= Scanner(System.out);
System.out.println("Enter the number:");
int n= Scanner.nextInt();
A.open();
```

2)
```
for(i<10; int = 1; i++)
System.in.println(n+"+"+i"="n+i);
```

Q1> A- Determine whether each of the following is true or false (**Answer 4 only**):
(4 marks)

1- One-dimensional array has two indices.
2- Method contains a set of instructions that perform well defined task and only runs when it called.
3- To access the elements of an array we use the [ ] operator.
4- Implicit type casting, a character variable can be truncated to an integer.
5- An array can store many different types of values.